

NATURAL FREQUENCIES OF IN-PLANE VIBRATION OF ANNULAR PLATES

The natural frequencies (the dimensionless frequency parameters) of in-plane vibration are tabulated for thin annular plates under practical combinations of boundary conditions. Though the frequencies of in-plane vibration are of practical importance in the design of piezoelectric disks, etc., there have not been very many papers available on them. Aggarwal [1], Kane and Mindlin [2], Gazis and Mindlin [3], Chen and Liu [4], and Ikegami *et al.* [5] studied the axisymmetric vibration of circular plates by using the analytical solutions to the vibration equations, and Holland [6] obtained a theoretical solution for the asymmetric vibration. Ambati *et al.* [7] analyzed the axisymmetric and asymmetric vibrations of circular or annular plates. However, all of these studies are confined to circular or annular plates with free edges, and sufficient engineering data have not been presented for plates with other boundary conditions, except for that in a paper of Srinivasan and Ramamurti [8] in which an annular plate with clamped inner edge and free outer edge is treated by using the finite element method.

The numerical calculations have been carried out by use of the transfer matrix method which is simple as an analytical process and also assures an accuracy sufficient for practical purposes [9, 10]. A brief explanation of the method is necessary in order for the results presented to be easily understood. With the outer and inner radii denoted by a and b , respectively, the polar co-ordinates (r, θ) are taken in the middle surface of an annular plate. The governing equations of in-plane vibration of the plate are expressed as

$$\begin{aligned} \frac{1}{r} \frac{\partial}{\partial r}(rN_r) + \frac{1}{r} \frac{\partial N_{\theta r}}{\partial \theta} - \frac{N_{\theta}}{r} + \rho h \omega^2 u &= 0, \\ \frac{1}{r} \frac{\partial}{\partial r}(rN_{r\theta}) + \frac{1}{r} \frac{\partial N_{\theta}}{\partial \theta} + \frac{N_{\theta r}}{r} + \rho h \omega^2 v &= 0, \end{aligned} \quad (1)$$

where ρ is the mass density, h is the plate thickness, and ω is the frequency in radians/second. The components of the in-plane force are given by

$$\begin{aligned} N_r &= D \left\{ \frac{\partial u}{\partial r} + \frac{\nu}{r} \left(u + \frac{\partial v}{\partial \theta} \right) \right\}, & N_{\theta} &= D \left\{ \frac{1}{r} \left(u + \frac{\partial v}{\partial \theta} \right) + \nu \frac{\partial u}{\partial r} \right\}, \\ N_{r\theta} &= N_{\theta r} = \frac{1-\nu}{2} D \left\{ \frac{\partial v}{\partial r} + \frac{1}{r} \left(\frac{\partial u}{\partial \theta} - v \right) \right\}, \end{aligned} \quad (2)$$

in terms of the displacements u, v in the radial and circumferential directions, respectively, where D is the extensional rigidity expressed as $D = Eh/(1 - \nu^2)$ by using the Young's modulus E , Poisson's ratio ν and the plate thickness.

For the free vibration of the plate, one can take

$$\begin{aligned} u &= a\bar{u} \cos n\theta, & v &= a\bar{v} \sin n\theta, \\ (N_r, N_{\theta}) &= D(\bar{N}_r, \bar{N}_{\theta}) \cos n\theta, & (N_{r\theta}, N_{\theta r}) &= D(\bar{N}_{r\theta}, \bar{N}_{\theta r}) \sin n\theta, \end{aligned} \quad (3)$$

where n is the circumferential wave number, and the overscored quantities $\bar{u}, \bar{v}, \dots$ are the respective dimensionless variables. For simplicity of the analysis, the following dimensionless parameters are also introduced:

$$\xi = r/a \quad \text{and} \quad \beta = b/a \quad (\text{radii ratio}), \quad \lambda^2 = \rho h a^2 \omega^2 / D \quad (\text{frequency parameter}). \quad (4)$$

Upon eliminating N_θ and $N_{\theta r}$ from equations (1)–(3), the equations of vibration can be written as the matrix differential equation

$$\frac{d}{d\xi} \begin{Bmatrix} \bar{u} \\ \bar{v} \\ \bar{N}_{r\theta} \\ \bar{N}_r \end{Bmatrix} = \begin{bmatrix} -\frac{\nu}{\xi} & -\frac{n\nu}{\xi} & 0 & 1 \\ \frac{n}{\xi} & \frac{1}{\xi} & \frac{2}{1-\nu} & 0 \\ \frac{n(1-\nu^2)}{\xi^2} & \frac{n^2(1-\nu^2)}{\xi^2} - \lambda^2 & -\frac{2}{\xi} & \frac{n\nu}{\xi} \\ \frac{1-\nu^2}{\xi^2} - \lambda^2 & \frac{n(1-\nu^2)}{\xi^2} & -\frac{n}{\xi} & -\frac{1-\nu}{\xi} \end{bmatrix} \begin{Bmatrix} \bar{u} \\ \bar{v} \\ \bar{N}_{r\theta} \\ \bar{N}_r \end{Bmatrix}. \quad (5)$$

Equation (5) can be written as

$$\frac{d}{d\xi} \{z(\xi)\} = [U(\xi)]\{z(\xi)\}, \quad (6)$$

by use of the state vector $\{z(\xi)\} = \{\bar{u}\bar{v}\bar{N}_{r\theta}\bar{N}_r\}^T$ and the coefficient matrix $[U(\xi)]$. In general, the vector $\{z(\xi)\}$ can be expressed as $\{z(\xi)\} = [T(\xi)]\{z(\beta)\}$ by using the transfer matrix $[T(\xi)]$ of the plate in the radial direction. The substitution of the expression for $\{z(\xi)\}$ into equation (6) yields

$$\frac{d}{d\xi} [T(\xi)] = [U(\xi)][T(\xi)]. \quad (7)$$

The elements of the matrix $[T(\xi)]$ are determined numerically by using the Runge–Kutta–Gill integration method.

The boundary conditions of the plate are written as

$$\bar{N}_r = \bar{N}_{r\theta} = 0 \quad \text{at a free edge}, \quad \bar{u} = \bar{v} = 0 \quad \text{at a clamped edge}, \quad (8)$$

The substitution of the expression $\{z(1)\} = [T(1)]\{z(\beta)\}$ into a given set of the boundary conditions (8) yields the respective frequency equations with the elements of $[T(1)]$ at the outer edge necessary for the calculation.

Table 1 presents the frequency equations of the plates under four combinations of free and clamped edge conditions, where the symbols (i) and (o) in the parentheses denote the inner and outer edges, respectively. The frequency parameters λ of the plates are determined by calculating the eigenvalues of the equations, and the natural frequencies

TABLE 1
Frequency equations of in-plane vibration of annular plates

Free(i)–free(o)	Free(i)–clamped(o)
$\begin{bmatrix} T_{31} & T_{32} \\ T_{41} & T_{42} \end{bmatrix}_{(1)} \begin{Bmatrix} \bar{u} \\ \bar{v} \end{Bmatrix}_{(\beta)} = 0$	$\begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix}_{(1)} \begin{Bmatrix} \bar{u} \\ \bar{v} \end{Bmatrix}_{(\beta)} = 0$
Clamped(i)–free(o)	Clamped(i)–clamped(o)
$\begin{bmatrix} T_{33} & T_{34} \\ T_{43} & T_{44} \end{bmatrix}_{(1)} \begin{Bmatrix} \bar{N}_{r\theta} \\ \bar{N}_r \end{Bmatrix}_{(\beta)} = 0$	$\begin{bmatrix} T_{13} & T_{14} \\ T_{23} & T_{24} \end{bmatrix}_{(1)} \begin{Bmatrix} \bar{N}_{r\theta} \\ \bar{N}_r \end{Bmatrix}_{(\beta)} = 0$

are obtained from

$$f = (\lambda/2\pi a)\sqrt{E/\rho(1-\nu^2)} \quad (\text{Hz}). \quad (9)$$

Tables 2-5 present the first two frequency parameters λ of thin annular plates for each of the wave numbers n under four combinations of boundary conditions. The vibration with $n=0$ is the axisymmetric one in which radial (extensional) or torsional vibration arises separately. The vibrations with $n=1, 2, 3, \dots$ are the asymmetric ones in which compound vibrations arise in the plates. In Tables 2 and 3, the plate with $\beta=0$ is a circular plate for which the frequency parameters have been obtained by taking an extremely small value for β , namely $\beta=0.01$. The first torsional vibration with $n=0$ and the first compound vibration with $n=1$ arising in the plates with free edges are the

TABLE 2
Frequency parameters λ of in-plane vibration of free(i)-free(o) annular plates; $\nu=0.3$

Radii ratio β	Circumferential wave number n					
	0	1	2	3	4	
	Radial	Torsional	Compound			
0	2.049†	3.039	1.618†	1.387†	2.131†	2.774†
Circular	5.388†	4.981	3.532†	2.511†	3.452†	4.401†
0.2	1.802	3.090	1.652	1.110	2.071	2.767
	4.748	5.209	3.842	2.403	3.401	4.389
0.4	1.452	3.530	1.683	0.721	1.618	2.482
	5.529	6.445	4.044	2.451	3.346	4.227
0.6	1.220	4.867	1.618	0.418	1.043	1.752
	7.979	9.409	5.092	2.474	3.433	4.386
0.8	1.065	9.380	1.488	0.178	0.487	0.892
	15.754	18.630	9.458	2.339	3.299	4.290

TABLE 3
Frequency parameters λ of in-plane vibration of free(i)-clamped(o) annular plates; $\nu=0.3$

Radii ratio β	Circumferential wave number n					
	0	1	2	3	4	
	Radial	Torsional	Compound			
0	3.831	2.267	1.958	3.046	3.964	4.777
Circular	7.013	4.151	3.180	4.085	5.024	5.977
0.2	3.303	2.284	2.104	2.553	3.688	4.712
	6.482	4.283	3.303	3.948	4.859	5.894
0.4	3.300	2.464	2.517	2.721	3.214	3.955
	8.086	5.127	3.508	4.147	4.998	5.874
0.6	4.306	3.084	3.260	3.630	3.942	4.250
	11.906	7.298	4.355	4.614	5.243	6.110
0.8	8.073	5.295	5.402	5.708	6.174	6.741
	23.635	14.182	8.074	8.080	8.106	8.181

TABLE 4

Frequency parameters λ of in-plane vibration of clamped(*i*)-free(*o*) annular plates; $\nu = 0.3$

Radii ratio β	Circumferential wave number n					
	0	1	2	3	4	
	Radial	Torsional	Compound			
0.2	2.204†	0.352	0.919†	1.542	2.157	2.778
	6.095	3.446	2.121	2.605	3.473	4.408
0.4	2.714†	0.823	1.281†	1.965	2.445	2.911
	7.922	4.522	2.691	2.908	3.604	4.492
0.6	3.924	1.662	1.952	2.608	3.294	3.722
	11.790	6.809	3.912	3.929	4.139	4.764
0.8	7.780	4.038	4.161	4.510	5.036	5.684
	23.539	13.761	7.774	7.760	7.748	7.750

TABLE 5

Frequency parameters λ of in-plane vibration of clamped(*i*)-clamped(*o*) annular plates; $\nu = 0.3$

Radii ratio β	Circumferential wave number n					
	0	1	2	3	4	
	Radial	Torsional	Compound			
0.2	4.236	2.506	2.783	3.378	4.066	4.802
	8.056	4.766	4.060	4.360	5.104	6.003
0.4	5.391	3.189	3.429	4.023	4.707	5.287
	10.558	6.246	5.306	5.311	5.619	6.289
0.6	7.930	4.692	4.835	5.237	5.832	6.541
	15.747	9.316	7.888	7.818	7.802	7.891
0.8	15.738	9.310	9.370	9.547	9.835	10.223
	31.431	18.595	15.721	15.676	15.617	15.557

rotation and the translation, respectively, corresponding to such modes for which the respective frequency parameters are zero. The values marked with † in Tables 2 and 4, respectively, have been compared with the results of Ambati *et al.* [7] and Srinivasan and Ramamurti [8] obtained by other methods. The results obtained by the present method are in good agreement with those of other authors within four significant figures, which indicates the accuracy of this method. The frequency parameters are largest for the clamped(*i*)-clamped(*o*) plates, and become smaller in that order for the free(*i*)-clamped(*o*) plates, the clamped(*i*)-free(*o*) plates, and the free(*i*)-free(*o*) plates. With an increase of the radii ratio β , the parameters monotonically increase except for the free(*i*)-free(*o*) plates.

One can obtain the natural frequencies of in-plane vibration of circular or annular plates with the given boundary conditions and dimensions directly from Tables 2-5, and can also easily obtain the values by the aforementioned process for the plates with dimensions not tabulated here.

The numerical computations presented here were carried out on a HITAC M-280H computer of the Hokkaido University Computing Center.

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REFERENCES

1. R. R. AGGARWAL 1952 *Journal of the Acoustical Society of America* **24**, 463-467. Axially symmetric vibrations of a finite isotropic disk. I.
2. T. R. KANE and R. D. MINDLIN 1956 *Journal of Applied Mechanics* **23**, 277-283. High-frequency extensional vibrations of plates.
3. D. C. GAZIS and R. D. MINDLIN 1960 *Journal of Applied Mechanics* **27**, 541-547. Extensional vibrations and waves in a circular disk and a semi-infinite plate.
4. S. S. H. CHEN and T. M. LIU 1975 *Journal of the Acoustical Society of America* **58**, 828-831. Extensional vibration of thin plates of various shapes.
5. S. IKEGAMI, T. NAGATA and Y. NAKAJIMA 1976 *Journal of the Acoustical Society of America* **60**, 113-116. Frequency spectra of extensional vibration in $\text{Pb}(\text{Zr} \cdot \text{Ti})\text{O}_3$ disks with Poisson's ratio larger than $1/3$.
6. R. HOLLAND 1966 *Journal of the Acoustical Society of America* **40**, 1051-1057. Numerical studies of elastic-disk contour modes lacking axial symmetry.
7. G. AMBATI, J. F. W. BELL and J. C. K. SHARP 1976 *Journal of Sound and Vibration* **47**, 415-432. In-plane vibrations of annular rings.
8. V. SRINIVASAN and V. RAMAMURTI 1979 *Journal of Design Engineering Paper No. 79-DET-100*, 1-4. In-plane vibration of annular disks using finite elements.
9. T. IRIE, G. YAMADA and S. AOMURA 1979 *Journal of Sound and Vibration* **66**, 187-197. Free vibration of a Mindlin annular plate of varying thickness.
10. T. IRIE, G. YAMADA and Y. MURAMOTO 1984 *Journal of Sound and Vibration* **95**, 31-39. Free vibration of joined conical-cylindrical shells.